

Module 2 Assessment- Planning Analysis Sheet

Liverpool FC Talent Pathway

Planning analysis sheet · UCD Professional Academy · Module 2 Python Assignment · Sean Phillips

1. Project Idea

A small Flask web app themed on Liverpool FC. A visitor reads about the club, fills in a player profile, is given an overall rating out of 100 calculated in Python, and is shown the contract offers that rating unlocks. They pick an offer and confirm it.

Goal	Show Python logic (data structures, functions, control flow, validation) driving a multi-page website.
Target user	A football fan who wants a quick, fun "would Liverpool sign me?" experience on phone or desktop.
Success looks like	User gets from the home page to a signed contract in three clear steps, with helpful errors if the form is wrong.

2. User Stories

#	As a visitor I want to...	Page
1	read about the club and its honours so I know what the site is about	Home
2	see the current squad and which shirt numbers are free	Squad
3	submit my player details and be told clearly if anything is wrong	Trials
4	see my rating, estimated value and the offers I qualify for	Offers
5	review an offer before committing, or go back and pick another	Contract (confirm)
6	get a confirmation once I sign	Contract (welcome)

3. Site Map and User Flow

Home (/) → Trials form (/trials) → if valid → Offers (/offers) → Confirm (/contract) → Welcome (/contract/confirm)

Squad /squad is reachable from the navigation bar on every page. Invalid form → the Trials page is shown again with error messages and the user's answers kept. Offers has a

"Change My Details" link back to Trials; Confirm has a "Look At Other Offers" link back to Offers.

4. Code Structure Plan

File	Responsibility
app.py	Flask routes only. Reads the form, calls the other modules, renders templates. Profile kept in the session between steps.
data.py	Club data as Python lists and dictionaries: honours, facts, squad, positions, feet, offers.
validation.py	validate_profile() checks every field and returns a cleaned profile plus a dictionary of errors.
ratings.py	calculate_rating(), estimate_market_value(), weekly_wage(), build_offers().
templates/	base.html (header, nav, footer) extended by index, squad, trials, offers, contract.
static/css/style.css	One stylesheet. Club red, responsive grid layouts, mobile first.

5. Form Inputs and Validation Rules

Field	Input type	Rule (checked on the server)
Full name	text	Required, max 60 characters
Age	number	Whole number, 16 to 40
Position	select	Must be one of the positions in data.py
Preferred foot	radio	Left, Right or Both
Shirt number	number	1 to 30 and not already taken by a squad player
Training days	number	1 to 7
Current club	text	Optional, max 60 characters

6. Rating Logic

Step	Plan
Base score	Everyone starts on 52
Age band	Points added by age band, highest for peak years 24 to 28 (+18)
Training	$(\text{training_days} - 3) * 3$, so fewer than 3 days costs points
Preferred foot	Both +6, Left +4, Right +2
Clamp	Final rating kept between 1 and 99
Market value	Grows with the square of the rating above 45, then an age multiplier
Offers	Each offer has a min_rating; only offers the rating reaches are shown

Security note: the chosen offer is sent in a hidden field, so the server rebuilds the offer list from the rating and only accepts a key that is in it. Editing the hidden field cannot unlock a better offer.

7. Design Decisions

Colour	Liverpool red as the main accent, dark text on light panels for contrast.
Typography	Anton for headings (bold, sporty), Inter for body text (readable).
Layout	Single column of panels; card grids use CSS grid auto-fit so they reflow on small screens.
Accessibility	Labels on every input, fieldset/legend for radios, table header scopes, error box uses role="alert".

8. Test Plan

Test	Expected result
Submit an empty form	Trials page reloads with an error for each required field
Age 15 or 41, shirt number 0 or 31	Range error shown, other answers kept in the form
Pick a shirt number held by a squad player	Error naming the number as taken
Low-rated profile vs high-rated profile	High rating sees more offers than low rating
Post a position that is not in the list	Rejected by server-side validation
Edit hidden offer_key to an offer not earned	Server refuses and returns the user to Offers
Open /offers without submitting a profile	Redirected to the Trials form
Resize to phone width	Grids stack to one column, no horizontal scroll

9. Deployment

Code on GitHub, one feature branch and pull request per step. Hosted on Render's free tier, auto-deploying from main. SECRET_KEY is set as an environment variable on Render, not stored in the repository.